

SQL Metrics & Aggregation Query Report

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Database System: MySQL Workbench

Target Schema: decode_labs_project3

1. Executive Summary

Following the successful data cleaning extraction phase completed in previous workflows, this technical report details the execution of structured SQL database queries on the verified e-commerce dataset table `decode_labs_dataset``. The scope of this phase involves validating schema structure, applying multi-level filtering criteria, calculating macro-level business KPI metrics, and generating targeted segmented matrices to drive strategic retail insights.

2. Query Implementation & Audit Trail

Step 1: Data Verification (Basic Retrieval)

Objective: Extract initial rows to confirm successful structure mapping and column sanity check across all clean vectors.

```
-- Retrieve the first 10 rows to verify all clean columns and metrics are present
SELECT *
FROM `decode_labs_dataset`
LIMIT 10;
```

	Cleaned_Order_ID	Cleaned_Date	Cleaned_Customer_ID	Cleaned_Products	Quantity	UnitPrice	Cleaned_Shipping_Address	Cleaned_Payment_Method	Cleaned_Order_Status	Cleaned_Tracking_Number	ItemsInCart	Cleaned_Coupon
▶	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	Save10
	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	Save10
	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	Freeship
	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	Save10
	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	Save10
	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	Save10
	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	6	Save10
	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	Freeship
	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2	Freeship
	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	Save10

Step 2: High-Volume Transaction Filtering (WHERE Clause)

Objective: Filter the entire dataset to isolate large orders where quantity thresholds meet or exceed 5 units.

```
-- Filter transactions where the product quantity sold is 5 or more
SELECT Cleaned_Order_ID, Cleaned_Customer_ID, Cleaned_Products, Quantity, UnitPrice
```

```
FROM `decode labs dataset`
WHERE Quantity >= 5;
```

	Cleaned_Order_ID	Cleaned_Customer_ID	Cleaned_Products	Quantity	UnitPrice
▶	ORD200000	C72649	Monitor	5	570.62
	ORD200002	C81728	Tablet	5	550.68
	ORD200007	C41460	Monitor	5	149.55
	ORD200010	C43443	Tablet	5	625.97
	ORD200019	C14552	Monitor	5	224
	ORD200020	C54829	Chair	5	62.51
	ORD200026	C21944	Laptop	5	59.55
	ORD200032	C12388	Tablet	5	536.72
	ORD200057	C88036	Printer	5	314.27
	ORD200065	C97735	Monitor	5	81.7
	ORD200066	C18104	Chair	5	127.18
	ORD200074	C25382	Phone	5	90.87

Step 3: Global Macro Aggregations (Total Orders & Revenue)

Objective: Compute primary corporate financial milestones capturing absolute sales volumes and operational metrics.

```
-- Calculate the total number of orders and the total overall revenue generated
SELECT
    COUNT(Cleaned_Order_ID) AS Total_Orders,
    SUM(Quantity * UnitPrice) AS Total_Revenue
FROM `decode labs dataset`;
```

	Total_Orders	Total_Revenue
▶	1200	1264761.9600000004

Step 4: Product Performance Breakdown (GROUP BY Segmentations)

Objective: Group individual records to analyze total unit movements and absolute revenue contribution sorted by high-velocity performers.

```
-- Group sales data by product to calculate total units sold and revenue per item
SELECT
    Cleaned_Products AS Product,
    SUM(Quantity) AS Total_Units_Sold,
    SUM(Quantity * UnitPrice) AS Revenue_Generated
FROM `decode labs dataset`
GROUP BY Product
ORDER BY Revenue_Generated DESC;
```

	Product	Total_Units_Sold	Revenue_Generated
▶	Chair	562	195620.110000000002
	Printer	542	195612.609999999996
	Laptop	535	192126.559999999994
	Tablet	497	186568.950000000007
	Monitor	480	175651.409999999997
	Desk	508	167459.93
	Phone	411	151722.39

Step 5: Transactional Basket Evaluation (AVG Analysis)

Objective: Calculate the baseline Average Order Value (AOV) across all completed transactions.

```
-- Calculate the average spending per transaction across the dataset
SELECT
    AVG(Quantity * UnitPrice) AS Average_Order_Value
FROM `decode labs dataset`;
```

	Average_Order_Value
▶	1053.96830000000005

Step 6: Customer Intelligence Analysis (Top Spenders Identification)

Objective: Uncover top purchasing profiles and VIP accounts by consolidating cumulative frequency and revenue matrices.

```
-- Identify Top Spenders (Customer Insights)
SELECT
    Cleaned_Customer_ID AS Customer_ID,
    COUNT(Cleaned_Order_ID) AS Total_Orders_Placed,
    SUM(Quantity * UnitPrice) AS Total_Amount_Spent
FROM `decode labs dataset`
GROUP BY Cleaned_Customer_ID
ORDER BY Total_Amount_Spent DESC
LIMIT 10;
```

	Customer_ID	Total_Orders_Placed	Total_Amount_Spent
▶	C38840	2	5723.23000000000005
	C57276	1	3456.39999999999996
	C67260	1	3390.79999999999997
	C13877	1	3384.9
	C18404	1	3370.2
	C16775	1	3353.75
	C65986	1	3352.4
	C47778	1	3334
	C59183	1	3322.55
	C25276	1	3313.89999999999996

Step 7: High-Value Threshold Analytics (HAVING Filtering)

Objective: Isolate high-performing product groups that cross a precise corporate limit threshold of \$5,000.

```
-- Conditional Analytics using HAVING (High-Value Products Only)
SELECT
    Cleaned_Products AS Product_Name,
    SUM(Quantity * UnitPrice) AS Total_Revenue
FROM `decode labs dataset`
GROUP BY Cleaned_Products
HAVING Total_Revenue > 5000
ORDER BY Total_Revenue DESC;
```

	Product_Name	Total_Revenue
▶	Chair	195620.110000000002
	Printer	195612.609999999996
	Laptop	192126.559999999994
	Tablet	186568.950000000007
	Monitor	175651.409999999997
	Desk	167459.93
	Phone	151722.39

Step 8: Multi-Level Ordering Strategy

Objective: Apply complex priority sorting to dissect patterns linking massive volume procurement with premium item costs.

```
-- Multi-Level Sorting (Quantity and Price Analysis)
SELECT Cleaned_Order_ID, Cleaned_Products, Quantity, UnitPrice
FROM `decode labs dataset`
ORDER BY Quantity DESC, UnitPrice DESC
LIMIT 20;
```

	Cleaned_Order_ID	Cleaned_Products	Quantity	UnitPrice
▶	ORD200789	Tablet	5	691.28
	ORD201122	Monitor	5	678.19
	ORD200632	Laptop	5	678.16
	ORD200469	Chair	5	676.98
	ORD200328	Tablet	5	674.04
	ORD200107	Printer	5	670.75
	ORD200326	Laptop	5	670.48
	ORD201065	Printer	5	666.8
	ORD201031	Phone	5	664.51
	ORD200463	Laptop	5	662.78
	ORD200361	Printer	5	659.85

3. Strategic Next Steps

- **Audit Integration:** All 8 steps successfully create an immutable execution pipeline validating backend dataset metrics against target parameters.

- **Dashboard Pipeline:** These pre-compiled aggregate queries serve as structural views or reference matrices for direct integration into front-end business intelligence platforms (Tableau / Power BI).